BIG DATA IN PRACTICE

From customer data to better decisions and measurable busine value

A practical case study in customer retention, insurance and commercial decision-making



THE BIG QUESTION

What decision gets better because of this data?

Lars Glade | 19 October 2026

Today’s question: when does data create value?

A practical lens — not a technical deep dive

1 Find the signal

What does the data actually show — and how reliable is it?

2 Explain the signal

What might be driving the pattern? What else could explain it?

3 Act and measure

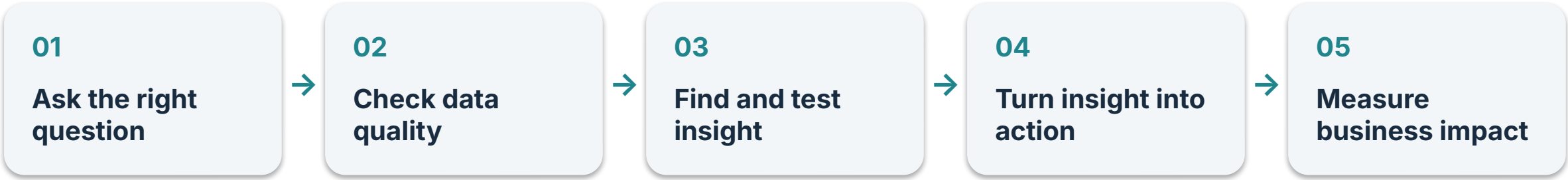
What decision changes, and how will we know it worked?

The goal is not to produce more reporting. It is to make better commercial decisions.

Examples are drawn from customer retention and insurance; the same logic applies across subscription businesses.

The journey from data to value

Five steps in a commercial analytics loop



At every step: challenge assumptions, document definitions, and involve the people who own the business result.

The 5 Vs — in practical terms

WHAT TO NOTICE

A checklist for deciding whether data is useful and trustworthy

VOLUME

Enough records?
Is the group large enough to compare?

VELOCITY

Current enough?
Will the data arrive in time to act?

VARIETY

Can sources connect?
Policies, claims, calls and campaigns.

VERACITY

Correct and complete?
What could be missing, duplicated or wrong?

VALUE

Which decision improves?
How will we measure the result?

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WHAT TO NOTICE
Data only matters when it changes a decision.

Start with the business question

Weak questions create impressive analysis with little value

Too vague

“Can we analyse churn?”

No decision, target group or outcome is specified.

Decision-ready

“Which customers are at risk, where can we intervene, and what incremental value can we retain?”

Before opening a dashboard, agree the decision, the metric, the time window and the owner.

Customer retention is a joined-up problem

WHAT TO NOTICE

One customer journey can leave traces in many systems

POLICIES

Products, premium, tenure

+

CLAIMS

Events, outcomes, cost

+

CONTACTS

Calls, complaints, effort

+

CAMPAIGNS

Audience, offer, response

+

FINANCE

Contribution, cost, value

Joined data → usable insight → a specific customer action → t

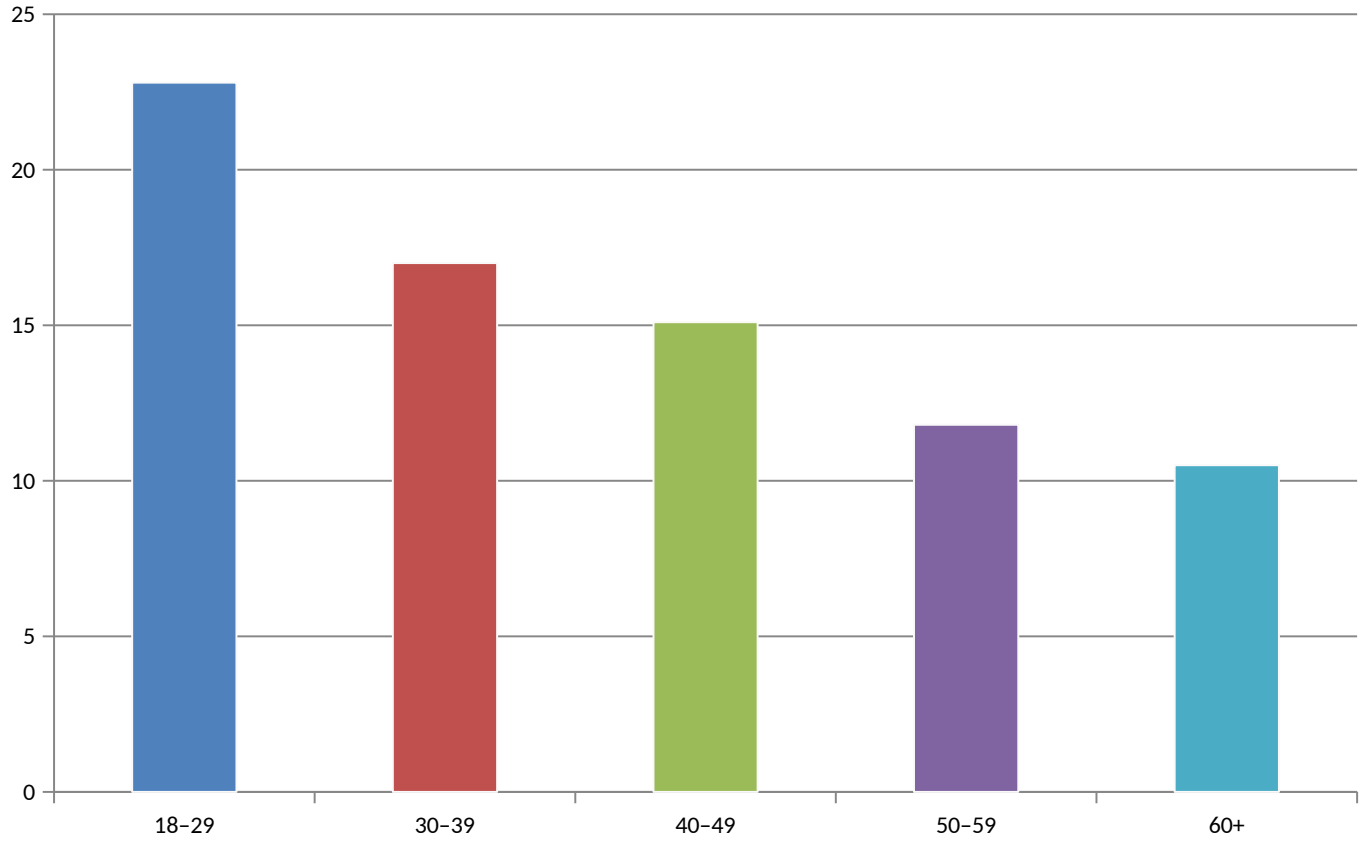


WHAT TO NOTICE

A missing contact record can make a good model look wrong.

Insight 1: churn is not evenly distributed

Historical insurance data showed higher churn among younger customers



Historical data shown in the original case; definitions and period should be verified before reuse.

What we observe

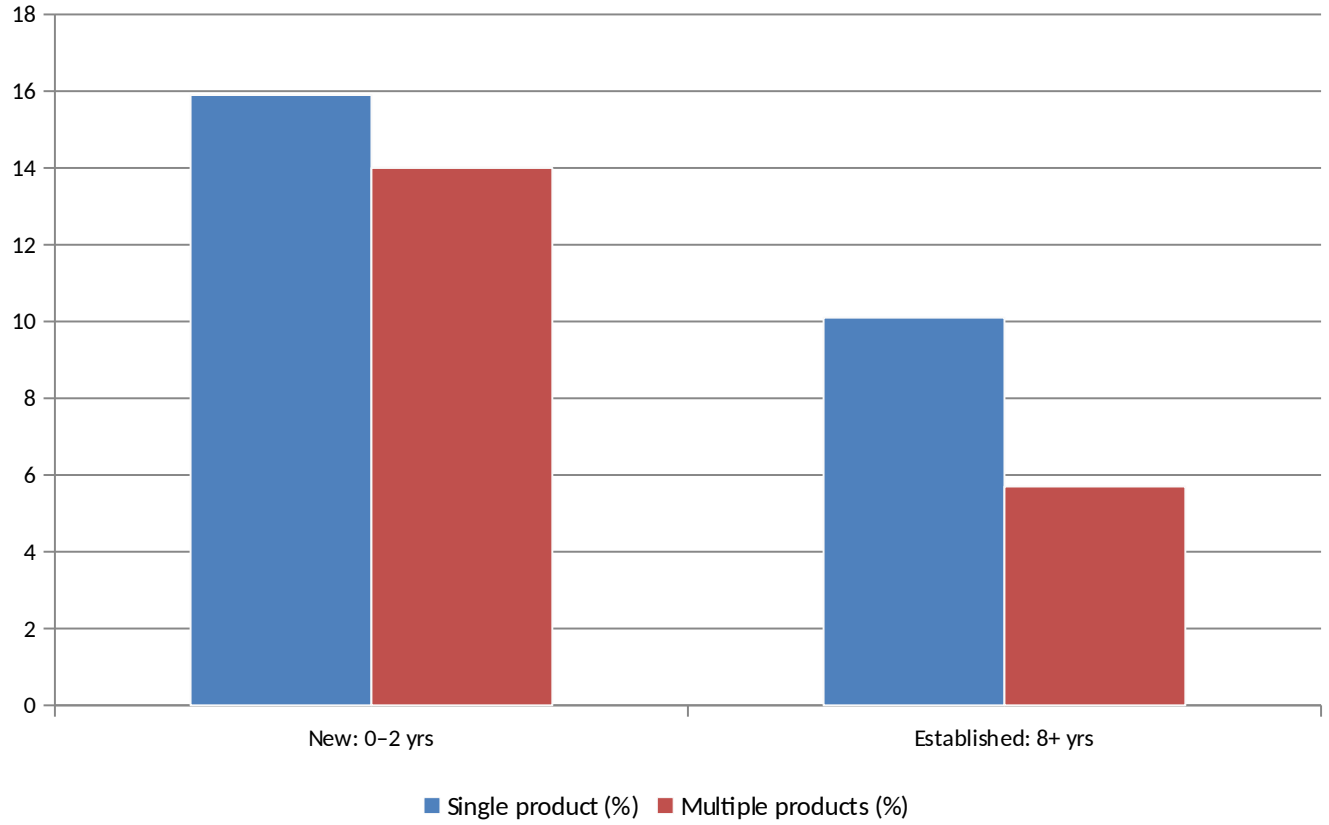
Reported annual churn declines across these age groups.

What we do not know yet

Age may proxy for tenure, life stage, price sensitivity or product needs.

Insight 2: more products, lower churn?

The relationship was stronger among established customers



New customers

Gap: 1.9 percentage points

Established customers

Gap: 4.4 percentage points

A difference is a clue, not proof that adding products causes loyalty.

A data difference is a clue, not an explanation

WHAT TO NOTICE

Move from observation to a testable hypothesis

01 DESCRIBE

What is the signal?

Confirm population, period and definition.

02 CHALLENGE

What else differs?

Price, age, tenure, need, channel or service experience.

03 TEST

What would prove it?

Compare similar customers; use a control group where possible.



Correlation can help us choose where to look. A fair test helps us learn whether caused a change.

WHAT TO NOTICE
Correlation is a clue, not proof.

Where should retention effort focus?

Customer tenure changes both risk and the economics of intervention

New customers

High churn risk; onboarding, expectation-setting and early service matter.

Established customers

Lower observed churn; protect relationship value and identify meaningful needs.

At-risk customers

Prioritise only when likelihood to leave, actionability and value justify cost.

A good model does not simply rank risk. It helps decide who to contact, when, with what offer — and when not to act.

Predictive modelling: useful, but not magic

A churn score is a prioritisation tool — not a customer truth

Highest risk / highest value	Personalised intervention
High risk / moderate value	Low-cost targeted contact
Lower risk or low expected value	Digital guidance or no action

Always ask

Does the score predict well? Can we act? Is the action profitable and fair?

Evaluate model quality, contact capacity, treatment effect and net value separately.

Customer value changes the decision

Risk alone is not enough to prioritise activity

LOWER VALUE

High churn risk ≠ automatic phone call

Use low-cost service or digital options where appropriate.

HIGHER VALUE

High churn risk + actionable opportunity

Consider a tailored intervention — if incremental value exceeds cost.

Prioritisation = risk × value × actionability, constrained by cost and customer experience.

Insight only matters when it changes the operating model

Connect analytics, frontline teams and commercial governance

DATA

Reliable and joined

→

INSIGHT

Interpreted with context

→

ACTION

Offer / channel / timing

→

LEARNING

Outcome and economics

Governance must make clear who owns definitions, data fixes, campaign
and results.



COMMERCIAL TEST

Did we retain value — or just create activity?

Reporting should explain movement — not just show it

A regular churn review turns metrics into management action

1 State of the union

What changed since the last review? Which segments and products moved?

2 Explain deviations

Separate known causes, hypotheses and data-quality issues.

3 Agree actions

Owner, next step, expected effect, deadline and follow-up metric.

Finance should be involved early — especially on metric definitions and attribution.



Where big-data work can go wrong

WHAT TO NOTICE

Small process errors can produce confident but wrong conclusions

Incorrect recording

Wrong customer or policy ID; campaign code entered incorrectly; invalid email or phone.

Missing information

No contact history; extra services not recorded; channel or interaction absent.

Broken customer journey

Customer moved without communication; dates do not align; duplicate identities.

Weak definitions & ownership

Different churn definitions; unclear metric owner; nobody accountable for fixing source data.



Before modelling: trace how the data was created, what is missing, and who can validate it.

ASK WHY

Is the pattern real — or is the data incomplete?

Other traps: the model can be right and the decision wrong

Data quality is necessary — business interpretation matters too

- Selection bias: only customers who contacted us are surveyed; quiet customers may be invisible.
- Reverse causality: loyal customers may buy more products — buying more may not create loyalty.
- Campaign contamination: treatment and control groups overlap or are exposed to other offers.
- Timing errors: outcomes are measured before the action has had time to work.
- Proxy variables: a score may encode past process bias rather than genuine customer need.
- Metric gaming: teams optimise the reported KPI instead of the underlying customer and business outcome.

Customer experience data: useful, but incomplete

WHAT TO NOTICE

NPS and CES are signals to investigate — not the full customer story

NPS

Would you recommend us?

A relationship / advocacy signal.

CES

How easy was it to get help?

An effort / friction signal.

Operational data

What happened, in which channel, at what cost — and what followed?



Combine what customers say with what they do and what it costs

WHAT TO NOTICE
Customer feedback needs behavioural context.

Drill down to the actionable driver

Aggregate scores can hide variation between service areas

Overall customer experience

Claims handling	Contact centre	Digital journey
Cycle time, clarity, outcome	Wait, transfer, resolution	Findability, completion, errors

The useful question is not only “where is the score low?” but “which fix changes the experience and outcome?”

Build the business case with Finance

Measure the incremental effect — not the activity total

1 Define the outcome

Who counts as retained, for how long, and at what contribution?

2 Estimate what changed

Use a fair baseline or control group; isolate incremental impact.

3 Calculate net value

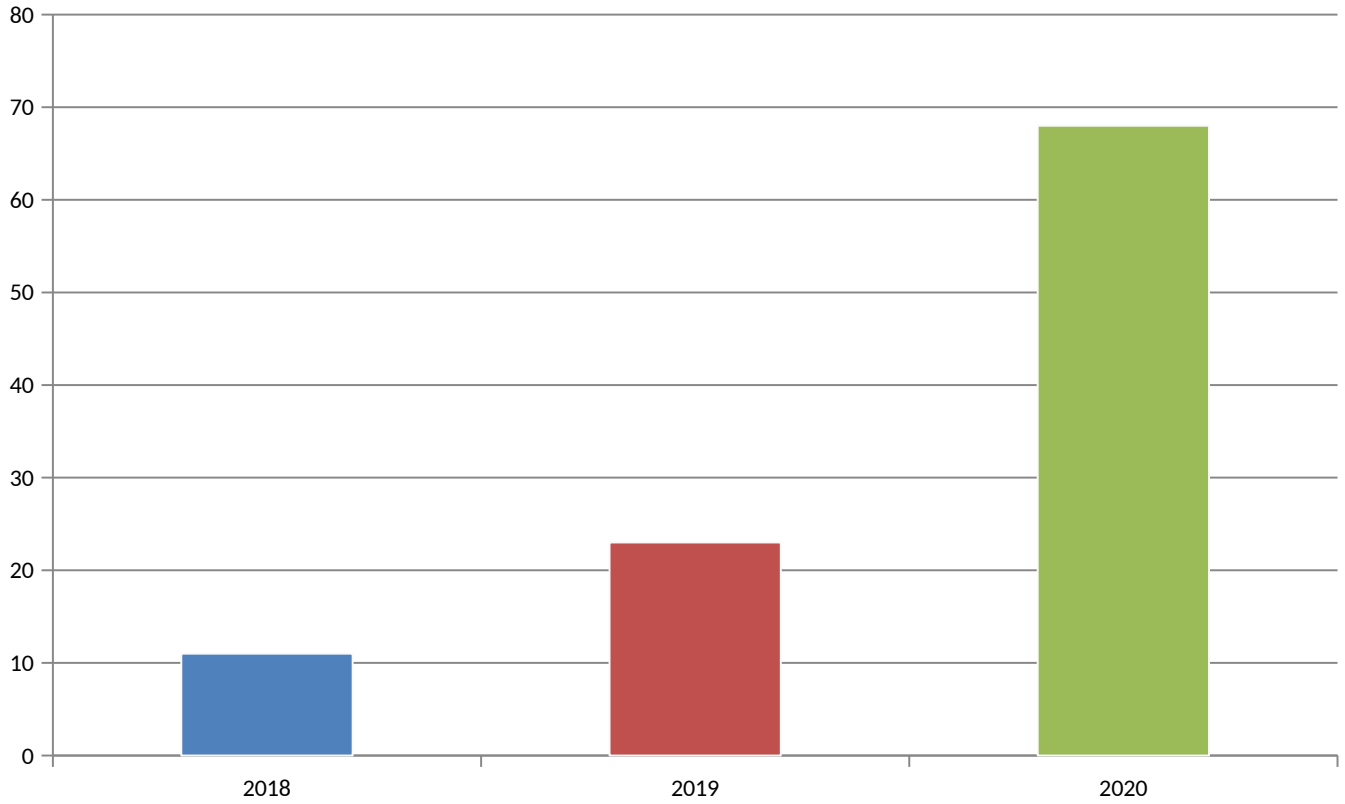
Incremental contribution retained – cost of intervention.

Net incremental value = additional customers retained × contribution margin – activity cost

Avoid double counting, gross-revenue claims and attributing all change to one initiative.

A reported trend is a starting point — not the whole business case

The original case tracked accumulated registered churn



Reported movement

From DKK 11m to DKK 68m over two years.

Before interpreting

Confirm scope, definition, timing, attribution and whether values are comparable.

Figures reproduced from the historical presentation; they require definition and context before being used externally.

Some customer events matter more than others

A car-policy cancellation could signal risk across the wider relationship

Car policy at risk → **Other policies exposed** → **Relationship value at risk**

The practical implication from the case: prevent the trigger where possible, rather than waiting to save every product after cancellation begins.

Test the hypothesis

Does this trigger predict wider churn after controlling for other factors?

Check the intervention

Can earlier service or a relevant offer reduce risk at an acceptable cost?

The human system is part of the data system

Good analytics depends on shared definitions and trusted relationships

Business

Own the decision, customer logic and action.

Analytics & data

Validate sources, methods, limitations and reproducibility.

Finance

Agree contribution, baselines, attribution and value.

Bring Finance in early. Bring frontline teams in early. Make ownership explicit.

If stakeholders do not trust the definitions, they will not trust the conclusion — even when the analysis is sound.

Mini-exercise: what would you check first?

WHAT TO NOTICE

A campaign reports lower churn among customers who received an offer

Observed result

Customers contacted by the retention team churned less over the next 90 days.

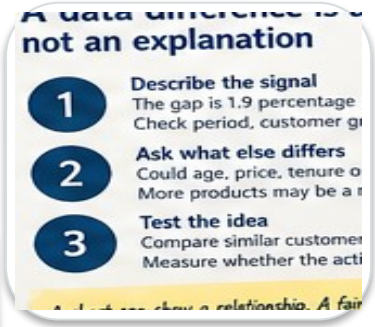
Possible explanations

The offer worked — or the team selected easier-to-retain customers, or contact records are incomplete.

Discuss in pairs:

- What data and definitions would you validate?
- What would make the comparison fair?
- Which business outcome would you measure next?

DISCUSS
What would you check before trusting the result?



Five habits of good commercial analytics

Take these into your next data project

- Start with a decision, not a dashboard.
- Check how the data was recorded and what is missing.
- Treat patterns as hypotheses; test competing explanations.
- Translate insight into a specific, owned action.
- Measure incremental value with Finance — including the cost.

DATA → INSIGHT → ACTION → MEASURABLE VALUE

Questions & discussion

Where have you seen data lead to the wrong conclusion — or a better decision?

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